

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examination-2023 Course: B. Tech. Semester : VI Branch: Mechanical & Allied Engg. Subject Code & Name: Renewable Energy Sources (BTMOE505B) Max Marks: 60 Date:10-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.			
		(Level/CO)	Marks
Q. 1	Describe the following.		
A)	Discuss various types of energy resources available on the Earth.	Understand	6
B)	Explain about the solar radiation received at the Earth's surface, mentioning spectral distribution of extra-terrestrial solar radiation and schematic representation of the mechanisms of absorption and scattering and beam and diffuse radiation received at the Earth's surface.	Remember/ Understand	6
Q.2	Attempt Any Two of the following.		
A)	Explain the main components of a flat – plate Solar collector with a neat Diagram. Also mention at least two basic points of differences between the Concentrating Collectors and Flat plate Collectors	Understand and Evaluate	6
B)	Explain the instruments used for the measurement of Solar radiation, with the help of neat schematic and its working principle.	Understand	6
C)	Describe about the single crystal Silicon Cell along with the principle of working of a Solar Cell, with neat schematics. Also list out the applications of PV system.	Understand /Synthesize	6
Q. 3	Describe Any Two of the following.		
A)	Describe the classification of wind turbine machines in terms of axis of rotation and rotor shapes. Also write down the expression for calculating wind power density.	Understand	6
B)	Explain the working of Ocean Thermal Energy conversion system with the help of neat schematic of components and cycle of operation.	Understand	6
C)	What are the environmental impacts of geothermal energy? What are the advantages of geothermal energy?	Understand	6
Q.4	Describe Any Two of the following.		
A)	What is Biomass? Discuss the various forms of biomass.	Understand	6

B)	Discuss the various parameters influencing the performance of a biogas plant.	Understand	6
C)	Discuss the devices used for harnessing the tidal energy, with the help of neat sketches.	Understand	6
Q. 5	Attempt any one of the following		
A)	What is a distillation process? Describe the construction and working of a Solar Still with the help of neat sketches	Understand	12
B)	Discuss construction and working of concentrating collector systems, with the help of neat schematic. Write down the advantages and disadvantages of concentrating collector systems.	Understand	12
	*** End ***		